SALARY VISUALIZATION

Analyzing the Impact of Demographic, Educational, and Employment Factors on Income Classification

Sector: Social Science

FACULTY - PROF. ARCHIT RAJ

Group 3

https://github.com/connectwithvanshika/C_G18_AirlineTicketPricesVsFuelCostsAnalysis

https://public.tableau.com/app/profile/syed.darain.qamar7769/viz/AirlinePricesAnalysis/ExecutiveOverview?publish=yes

TEAM MEMBERS

Atharva Tiwari
Rachit Gupta
Agrima Gusain
Tanmay Singh
R Ashrith
Ashish kushwaha

CONTEXT AND PROBLEM STATEMENT

SECTOR CONTEXT

Income levels are influenced by socio-economic factors such as education, occupation, workclass, demographics, and financial assets.

Understanding these patterns helps identify income inequality, economic mobility, and the factors that increase the likelihood of earning above $50K.

Key decision-makers:

- Policy makers → Design education and employment policies
- Workforce planners → Identify skill gaps and high-income occupations
- Social researchers → Analyze income inequality and demographic patterns

PROBLEM STATEMENT

Income inequality is influenced by factors such as education, occupation, workclass, demographics, and financial conditions.The real challenge is identifying which socio-economic factors most strongly increase the likelihood of earning above $50K.

Core Question:
Which demographic, educational, occupational, and financial factors influence high-income status?

And how can these insights support workforce planning, policy targeting, and economic mobility?

OBJECTIVE

The main goals of this analysis are:

- Measure how education, occupation, workclass, demographics, working hours, and capital gains affect the likelihood of earning above $50K
- Understand other influencing factors, such as:
  - Education level
  - Occupation type
  - Marital status
  - Gender
  - Capital gain and capital loss

DATA ENGINEERING (SOURCE → PIPELINE)

Source

- Dataset: Adult Census Income Dataset
- Size: 15,000+ records with 15 features
- Time Period: Income category (<=50K or >50K)

Python ETL (Cleaning Steps)

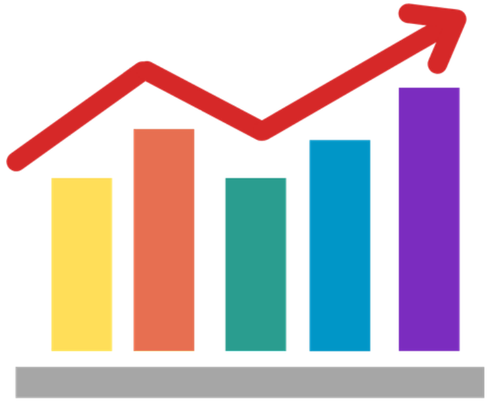
Handled missing values, standardized categorical values, grouped similar categories, and flagged outliers using the IQR method while retaining them because they represent meaningful real-world income variations such as extreme working hours, capital gains, and capital losses.

GitHub (Pipeline Work)

- Notebook 01: Data extraction and cleaning of the Adult Census Income dataset
- Notebook 02: Feature engineering for income analysis, including age groups, education groups, occupation groups, and income labels
- Final Output: Cleaned and transformed dataset ready for Tableau dashboarding and statistical analysis

Data Dictionary (Key Columns)

income	Income category (<=50K or >50K)
education_num	Numeric education level
occupation	Job role of the individual
hours_per_week	Weekly working hours
capital_gain	Profit from investments or asset sales



KPI & METRICS FRAMEWORK

High Income Rate
24.0%

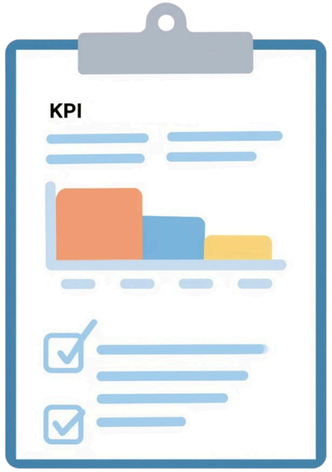
People with Capital Gains
8.7%

Gender Income Gap
18.9%

Top Earning Occupation
Exec-managerial: 45.9%

Why these KPIs?

- Directly aligned with the core problem of identifying factors that influence high-income status
- Help analyze income differences across gender, occupation, and financial indicators
- Capture the impact of socio-economic factors such as employment type, demographics, and capital gains on wealth patterns



High Income Rate

People with Captital Gains

Gender Income Gap

Top Earning Occupation

How are they computed?

- Calculated using Python on the cleaned Adult Census Income dataset
- Key computations:
  - High Income Rate → percentage of individuals earning >50K
  - Gender Income Gap → difference in high-income rate between males and females
  - Top Earning Occupation → occupation with the highest percentage of >50K earners
  - People with Capital Gains → percentage of individuals with capital_gain > 0
- Implemented in the analysis notebook pipeline

KEY INSIGHTS (EDA)

Higher education increases the likelihood of earning above $50K, showing that education is a strong income driver

Occupation plays a major role in income level, with executive and professional roles having the highest high-income rates

High-income patterns vary across demographic and job groups, showing the role of social and career factors.

Education, occupation, and capital gains influence high-income status more strongly than working hours alone

Income is strongly linked to education, occupation, and capital gains

Financial indicators like capital gain explain income differences more clearly than basic work-effort measures such as weekly working hours

Gender and marital status strongly influence earning potential.

Conflict phases (e.g., war periods) show higher fares and wider price variation, highlighting the impact of global instability

Higher education and professional roles are more strongly linked to high-income status.

Flights with longer distances show higher surcharge ratios and greater price fluctuations, making them more vulnerable to fuel shocks

Income is not driven by one factor alone; it is shaped by education, occupation, demographics, workclass, and financial indicators.

ADVANCED ANALYSIS

What advanced analysis was performed?

Correlation Analysis

- Checked links between income, education, hours worked, and capital gains.
- Result: Education and capital gains showed stronger association with high income than work hours.
- Insight: Income depends more on structural and financial factors than hours worked.

Hypothesis Testing

- Tested income differences across education, gender, and marital status.
- Result: Significant differences were found across major socio-economic groups.
- Insight: These factors meaningfully affect income distribution.

Prediction Analysis

- Estimated the chance of earning above $50K.
- Result: Higher education, professional roles, self-employment, and capital gains increased high-income probability.
- Insight: High-income status can be predicted using combined socio-economic factors.

Segmentation Analysis

- Grouped individuals based on income-related characteristics.
- Identified segments:
  - Low-income group
  - Mid-income potential group
  - High-income profile group
- Insight: Different socio-economic groups show distinct income patterns and earning potential.

Tools & Libraries Used

Pandas

NumPy

seaborn

Matplotlib

SciPy

scikit learn contrib

TABLEAU DASHBOARD WALKTHROUGH

Income Insights Dashboard

Provides a high-level overview of income patterns and socio-economic differences

- Key KPIs: High Income Rate, Gender Income Gap, Top Earning Occupation, People with Capital Gains
- Compares income across gender, education, occupation, workclass, age, and marital status
- Helps identify which groups are more likely to earn above $50K

Helps identify income patterns, inequality, and high-income drivers.

Dashboard Access

https://public.tableau.com/app/profile/agrima.gusain5797/viz/IncomePredictionAnalysisDashboard_17773823940220/Dashboard3

Operational View (For Analysts)

Enables detailed income analysis and decision support

- Income breakdown: education, occupation, gender, and marital status
- Workforce insights: workclass, working hours, and occupation groups
- Financial impact analysis: capital gain, capital loss, and high-income probability
- Demographic analysis: age group, sex, race, and native country

Helps analysts understand income drivers and socio-economic patterns in detail.

Interactivity (Live Filtering)

Education Group

Age Group

Income Label

Occupation Group

RECOMMENDATIONS

1. Education strongly improves income potential

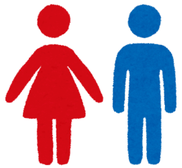
Insight: Higher education groups show higher chances of earning above $50K
Recommendation: Invest in education access and upskilling programs
Expected Impact: Improved economic mobility and higher earning potential

2. Occupation strongly affects income level

Insight: Executive and professional roles have the highest high-income rates
Recommendation: Focus skill development on high-paying career paths
Expected Impact: Better workforce readiness and improved income outcomes

3. Capital gains are a strong wealth signal

Insight: People with capital gains have a higher probability of earning above $50K
Recommendation: Promote financial literacy and investment awareness
Expected Impact: Improved wealth-building opportunities



4. Gender and marital income gaps are visible

Insight: Income varies across gender and marital-status groups
Recommendation: Support fair pay policies and targeted workforce inclusion programs
Expected Impact: Reduced income inequality

5. Working more hours does not guarantee high income

Insight: Weekly work hours have weaker impact than education, occupation, and capital gains
Recommendation: Prioritize skill quality and career mobility over longer work hours
Expected Impact: More productive and sustainable income growth

IMPACT & VALUE

Social Impact (So What?)

Income Mobility (+Improvement Potential)

- Education and upskilling can increase the share of high-income individuals
- Improves long-term earning potential

Workforce Development (+Career Growth)

- Training for high-paying occupations like executive, professional, and technical
- Helps move workers toward better income groups

Inequality Reduction (↓ Income Gap)

- Gender and marital income gaps highlight unequal earning patterns
- Supports fair pay policies and targeted inclusion programs

Financial Growth (+Wealth Building)

- Capital gains strongly increase high-income probability
- Promotes financial literacy and investment awareness

Policy Efficiency (+Better Targeting)

- Income patterns help identify groups needing support
- Improves effectiveness of education, workforce, and social policies

Cost of Inaction

- Income gaps remain unaddressed across gender, education, and occupation groups
- Low-income workers may miss skill-development and career-growth opportunities
- Policy decisions may be based on assumptions instead of data-backed evidence
- Limited financial awareness may reduce wealth-building opportunities

Why stakeholders should act

Income inequality is a major socio-economic challenge, and data-driven workforce, education, and financial policies can improve economic mobility and reduce income gaps.

LIMITATIONS & NEXT STEPS

Limitations

Limited explanatory power of model:
Income is influenced by many hidden factors, so the model cannot explain every income outcome

Missing key variables
Data does not include family background, location cost, industry type, experience, or personal wealth

External factors not fully captured:
Economic conditions, labor market trends, and policy changes are not directly included

Dataset limitation:
Analysis is based on census records and may not capture all modern income patterns

Future Scope

Build advanced predictive models
Use machine learning models to improve high-income prediction accuracy

Incorporate richer datasets
Add location, experience, industry, family income, and cost-of-living data

Apply advanced segmentation
Cluster individuals by education, occupation, workclass, and financial characteristics

Use causal analysis techniques
Move beyond correlation to understand the true drivers of high-income status

TEAM & CONTRIBUTIONS

Atharva Tiwari (Project Lead, PPT & Quality)

Led overall project direction, ensured quality, handled final presentation & reporting

Tanmay Singh (Data Lead)

Managed dataset sourcing and initial data structuring

R Ashrith (ETL Lead)

Built data pipeline, performed cleaning, preprocessing and Conducted EDA

Rachit Gupta (Analysis Lead)

Statistical analysis, and insight generation

Agrima Gusain (Visualization Lead)

Designed and developed Tableau dashboard

Ashish kushwaha (Strategy Lead)

Worked on business insights, recommendations, and impact analysis